

Probability and Uncertainty in Neural Modeling

Anna Freund

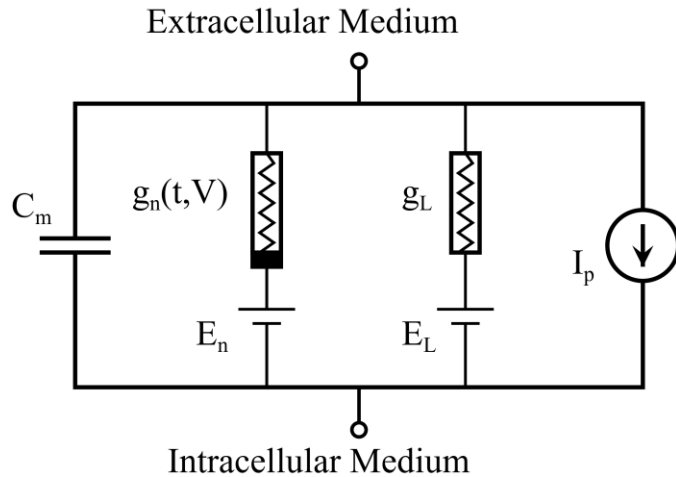
Computational Neuroscience Unit, OIST



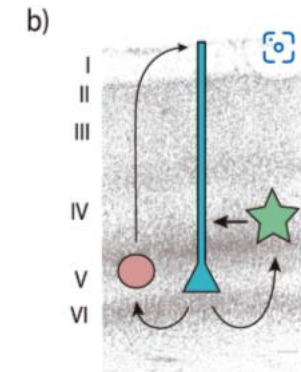
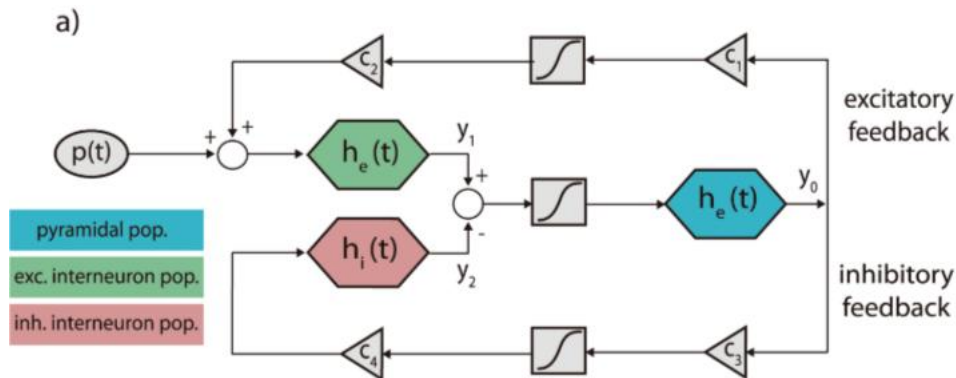
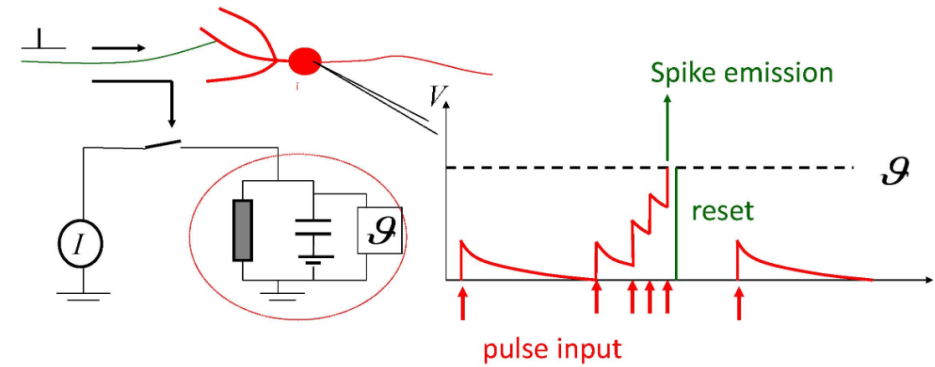
Contents

- Why do we need probabilities
- Basics of probability theory
- Random variables
- Probability distributions
- Variability in spike trains
- Poisson spike generator
- Noise
- Neural Data Analysis

Why probability in neural modeling?



Leaky Integrate-and-Fire Model



Deterministic model

Same input → Same output

Determinism is the metaphysical view that all events within the universe can occur only in one possible way



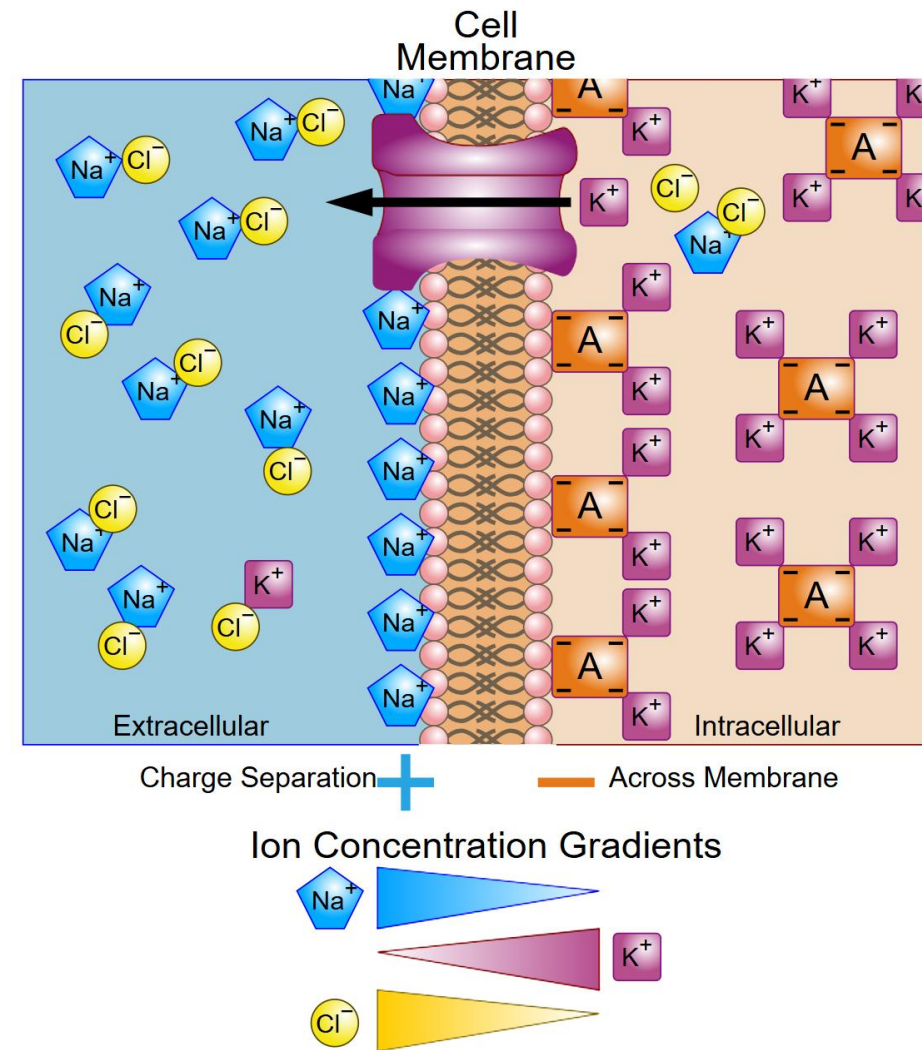
Why probability at all?

Scientists have established laws and equations to describe our surroundings

- Laws of Thermodynamics
- Newton`s Laws
- Membrane potential of cells

Behavior of macroscopic systems, leaving out properties of single molecules

We need to include the stochasticity of single molecules to model variability



Wikipedia: Membrane potential

Why probability in neural modeling?

Same input → output?

Single neurons and neural networks can be described with deterministic equations to model macroscopic events. Stochastic properties of single molecules are neglected.

- But we can see the variability in our measurements
- We want to understand and predict it

Variability in the brain

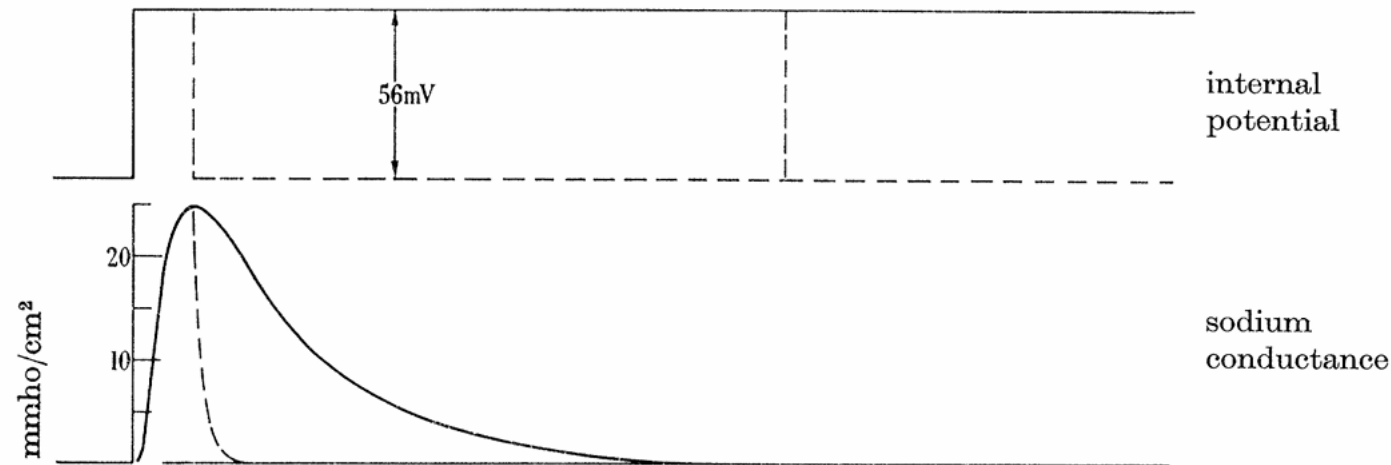
Same input → Variable output

- Spike trains of individual neurons are irregular
- If we would listen to measured electrical activity of a single neuron, we would only hear noise
- Neurons exhibit spontaneous activity, not related to stimuli
- High trial to trial variability in electrical recording with same stimulus
- Synaptic transmission is determined by probabilities
- Thermal energy: molecules move in along each dimension (kinetic energy)

We use probability to include variability and uncertainty

Stochastic openings of sodium channels

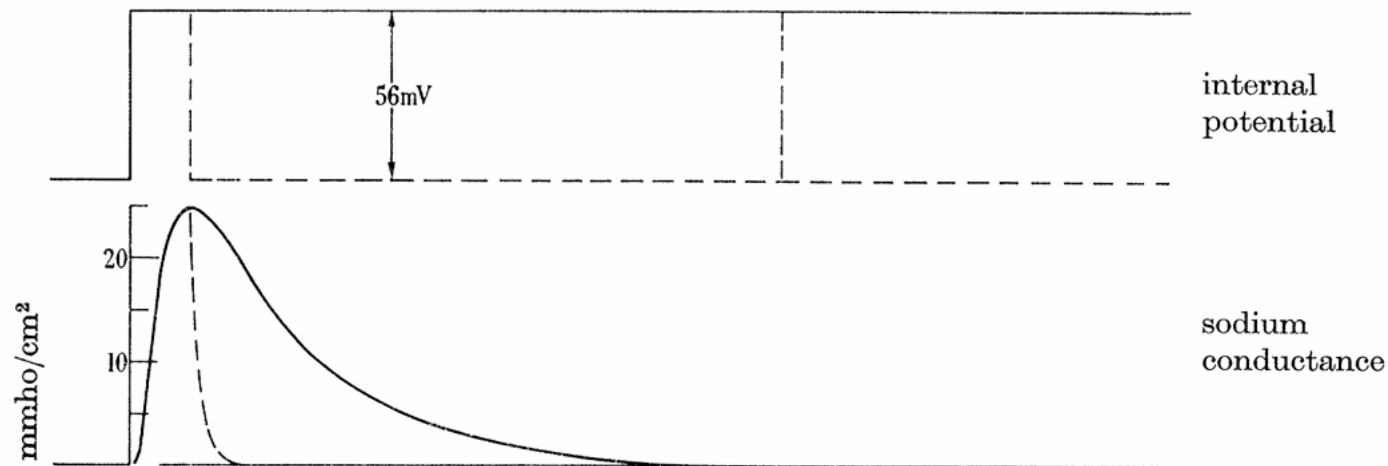
Hodgkin and Huxley described dynamics of membrane voltage in terms of deterministic ionic currents



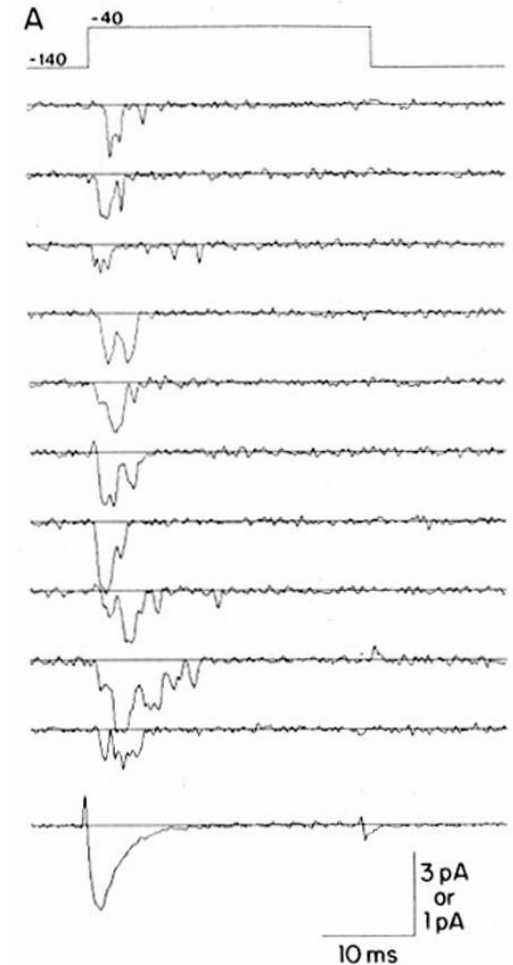
Hodkin(1958), JSTOR, The Croonian Lecture: Ionic Movements and Electrical Activity in Giant Nerve Fibers

Stochastic openings of sodium channels

- Patlak and Oriz recorded sodium channels from rat heart
- Currents are caused by stochastic molecular pores
- Macroscopic current is sum over microscopic currents of many channels



Hodkin(1958), JSTOR, The Croonian Lecture: Ionic Movements and Electrical Activity in Giant Nerve Fibers

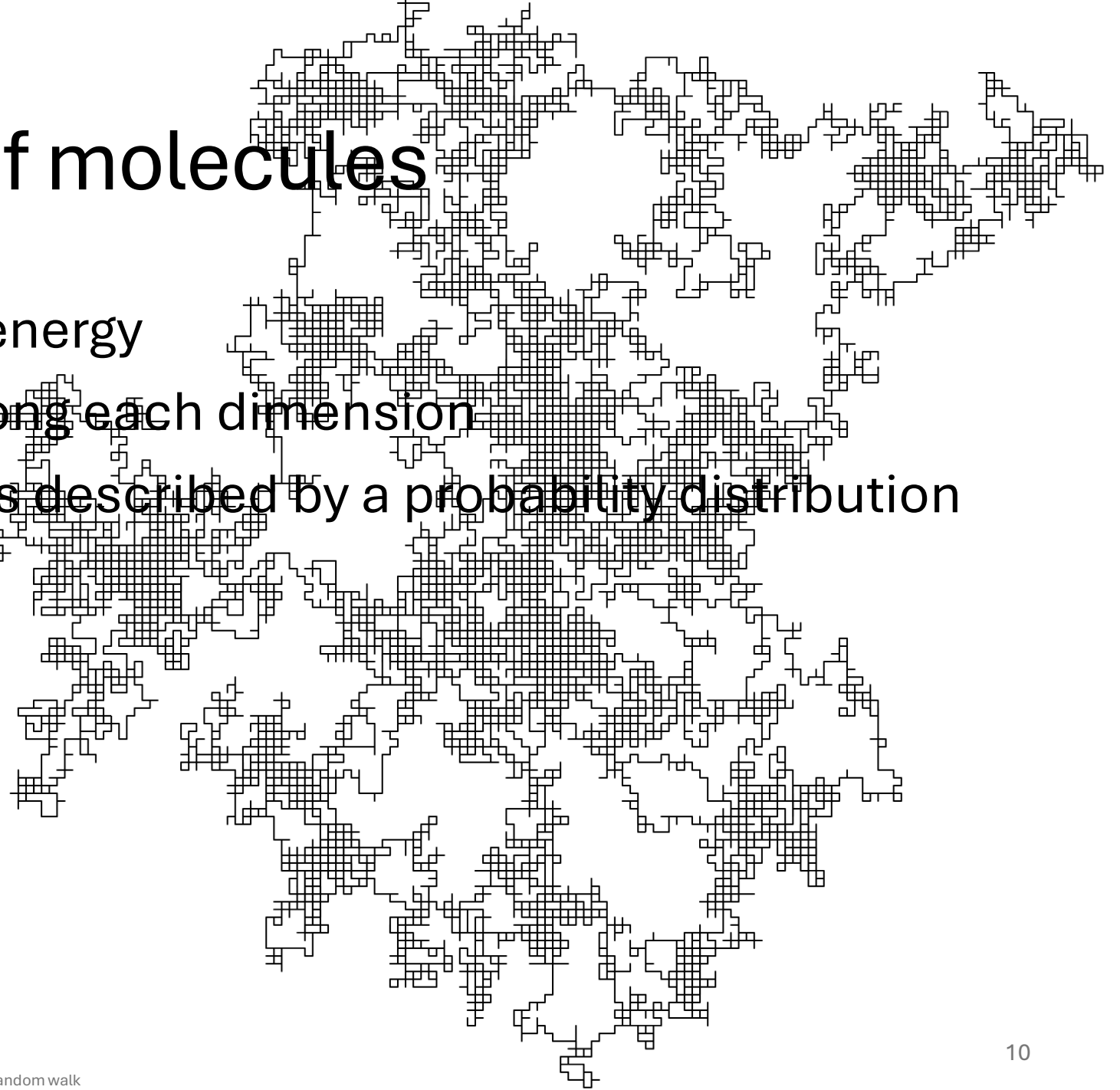


Patlak and Oriz (1985), Journal of General Physiology, Slow Currents Through Single Sodium Channels of the Adult Rat Heart

Thermal motion of molecules

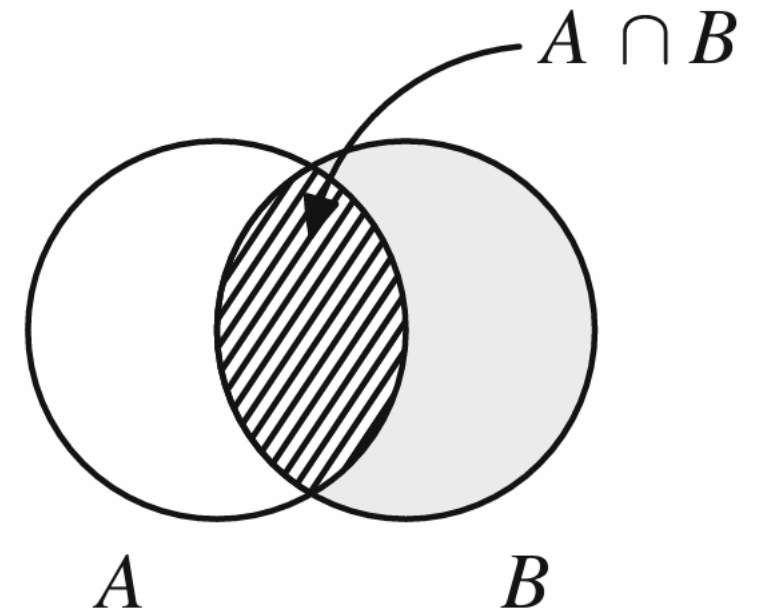
- Molecules have kinetic energy
- Results in movement along each dimension
- Direction of movement is described by a probability distribution

Random walk



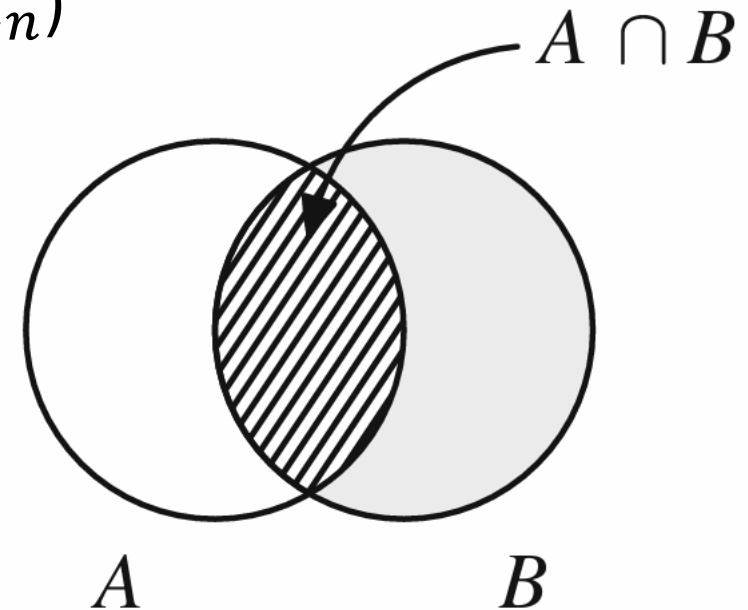
Probability

- Events ... outcomes from experiments or observations
- $P(A)$... probability of the event $A \rightarrow$ how likely A occurs
- Sample space Ω ... collection of all possible outcomes
- Union ... $A \cup B$
- Intersection ... $A \cap B$
- Complement ... A^C , not A
- Mutually exclusive ... empty intersection



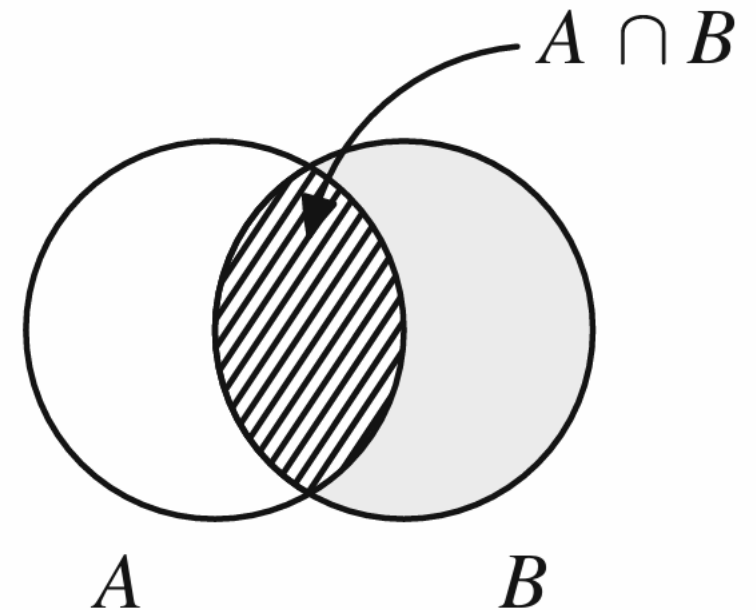
Axioms of Probability

- For all events A , $P(A) \geq 0$
- $P(\Omega)=1$
- If $A_1, A_2, \dots, A_n$ are mutually exclusive events, then
$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$



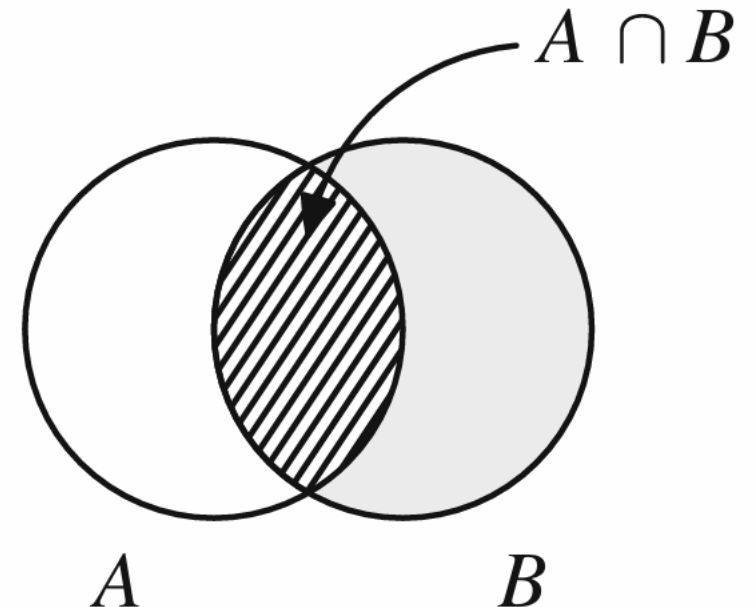
Properties of Probability

- $P(A^C) = 1 - P(A)$
- If A and B are mutually exclusive, $P(A \cap B) = 0$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



Conditional Probability

- $P(A|B)$... probability of A given B
- We limit the universe to B
- Take relative probability assigned to part of A this is contained in B
- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- $\rightarrow$ Multiplication rule: $P(A \cap B) = P(A|B) * P(B)$



Random variables

- Mathematical objects
- Describe variation among quantitative measurements
- Assign number to each outcome
- E.g., spike count of each trial
- Can be visualized by a histogram

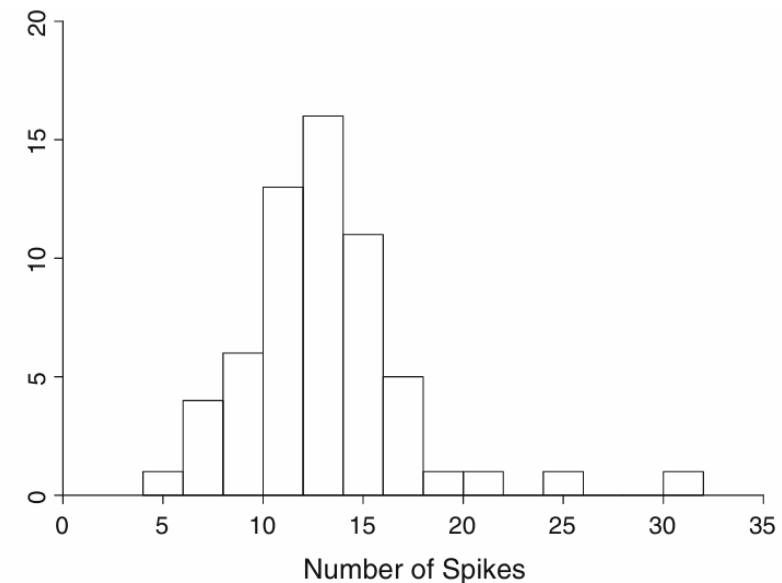


Fig. 3.2 Histogram of spike counts from a motor cortical neuron. The histogram displays 60 spike counts from a particular neuron recorded in primary motor cortex across 60 repetitions of the practiced condition.

Spike count from a motor cortical neuron

- To study cortical correlates of practicing a movement
- Recorded from primary motor cortex
- Button-pressing task
- Spike counts from single neuron
- 60 repetitions of same condition
- Shows considerable variation
- Uncertainty in random variable (spike count) is described by its probability distribution

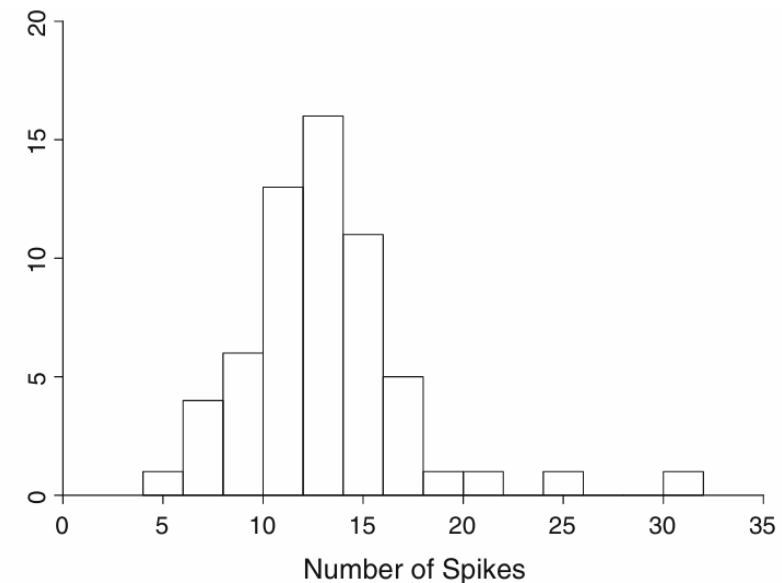


Fig. 3.2 Histogram of spike counts from a motor cortical neuron. The histogram displays 60 spike counts from a particular neuron recorded in primary motor cortex across 60 repetitions of the practiced condition.

Random variables

- Described by distributions
- $f(x)=P(X=x)$... probability mass function or probability density function (continuous random variable)
- $F(X)=P(X\leq x)$... cumulative distribution function
- Means and variances of the random variables can be computed from the distribution

Expectation (mean) of random variable x

$$\mu_X = E(X) = \sum_X x * f(x)$$

- μ_X ... *expected value*
- x ... *random variable*
- X ... *set of all events*
- $f(x)$... *probability of random variable having value x*

Variance of random variable x

$$\sigma_X^2 = V(X) = \sum_X (x - \mu_X)^2 * f(x)$$

- σ_X ... *standard deviation*
- μ_X ... *expected value*
- x ... *random variable*
- X ... *set of all events*
- $f(x)$... *probability of random variable having value x*

Discrete vs. continuous variables

- So far, we talked about discrete random variables
Spike count could have values 0,1,2,...,35
- For other measures like spike time interval, it would be more appropriate to consider all values within a defined interval
- X ... continuous random variable on interval (A,B) , with $A = -\infty$ and $B = \infty$

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

$$\int_A^B f(x)dx = 1, P(X = a) = 0$$

Probability distributions

- Describe how probability is distributed over the possible values of a random variable
- Different types of random variables
→ different types of distributions

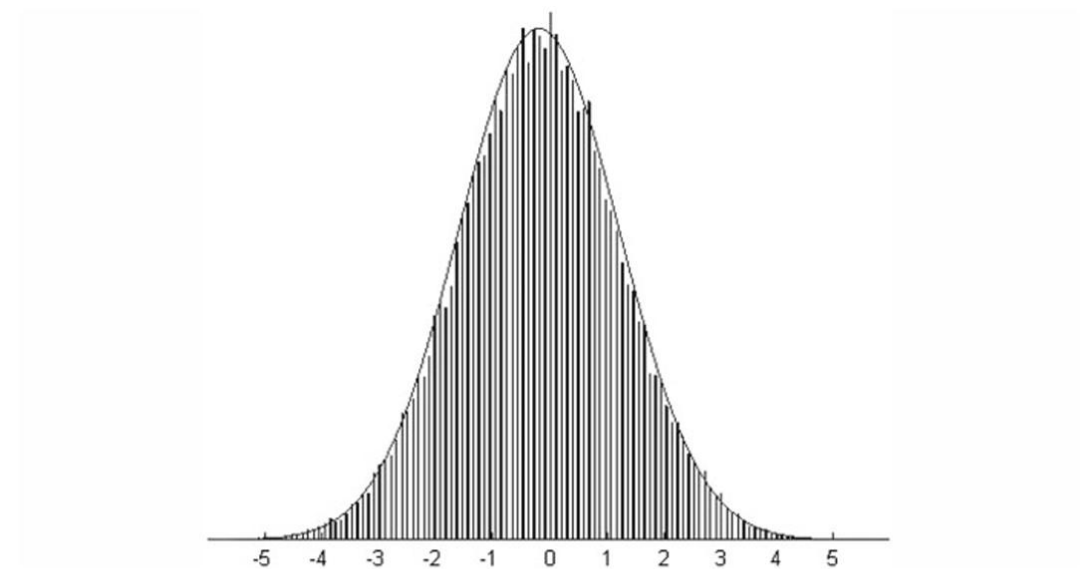
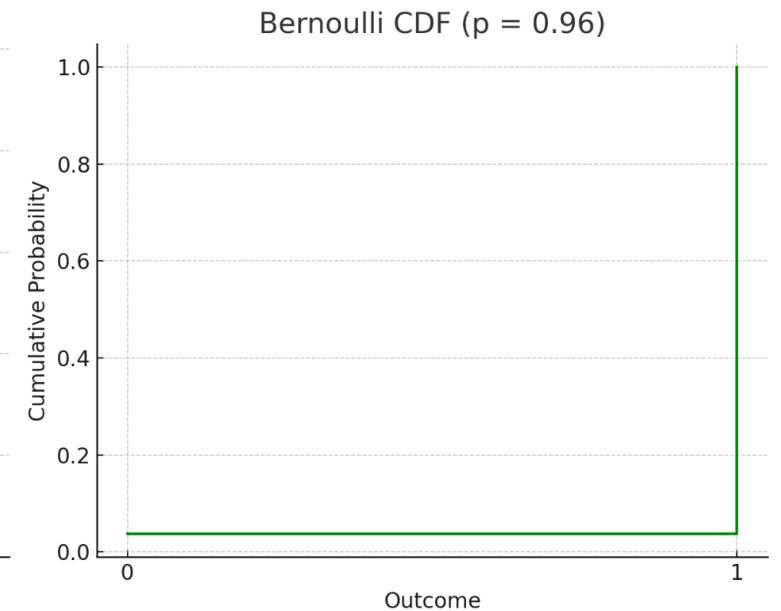
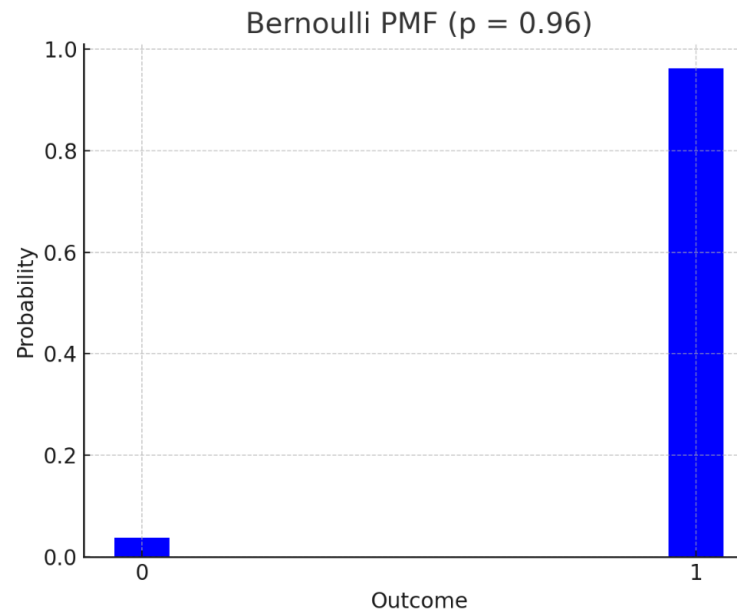


Fig. 3.4 A histogram of MEG noise at a SQUID sensor, overlaid with a normal density function (the “bell-shaped curve”).

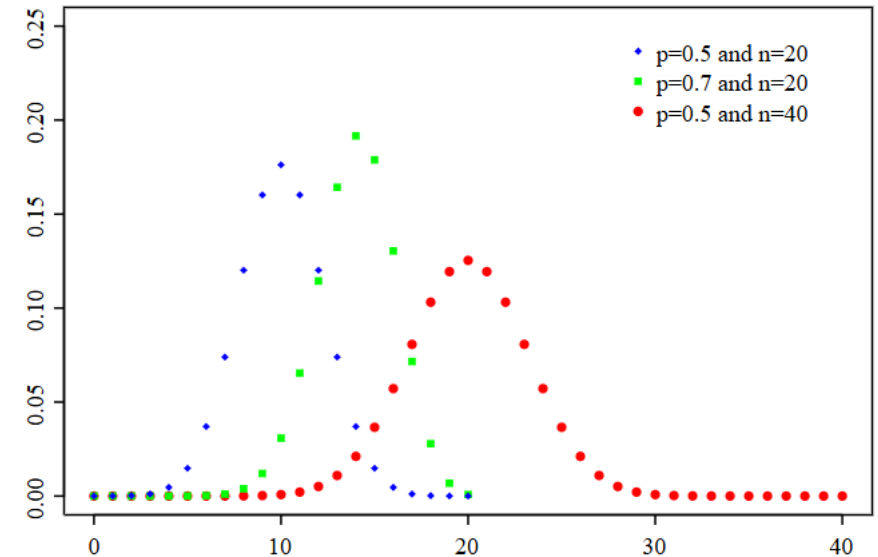
Bernoulli distribution

- Random variables takes values 0 or 1
- $P(X=1) = p$
- $P(X=0) = 1-p$
- $E(X) = p$
- $\text{Var}(X) = p^*(1-p)$



Binomial distribution

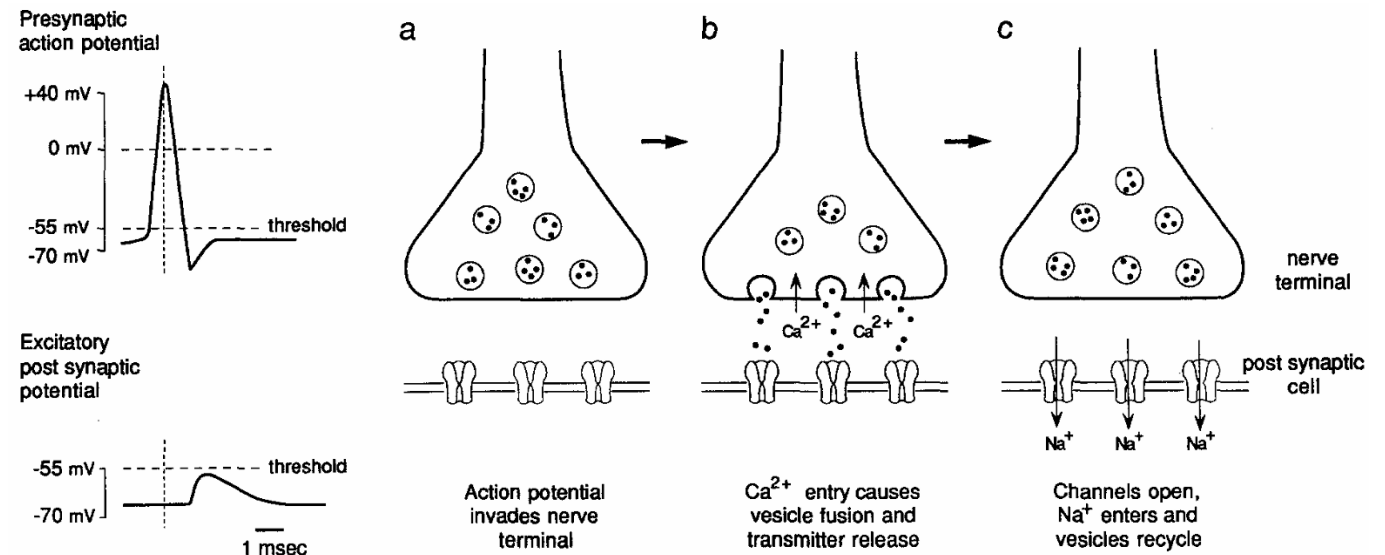
- Discrete
- Number of successes in a sequence of n independent experiments
- Probability of success: p
- Single experiment: Bernoulli trial
- $P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$
- $E(X) = n \cdot p$
- $\text{Var}(X) = n \cdot p \cdot (1-p)$



Quantal response in synaptic transmission

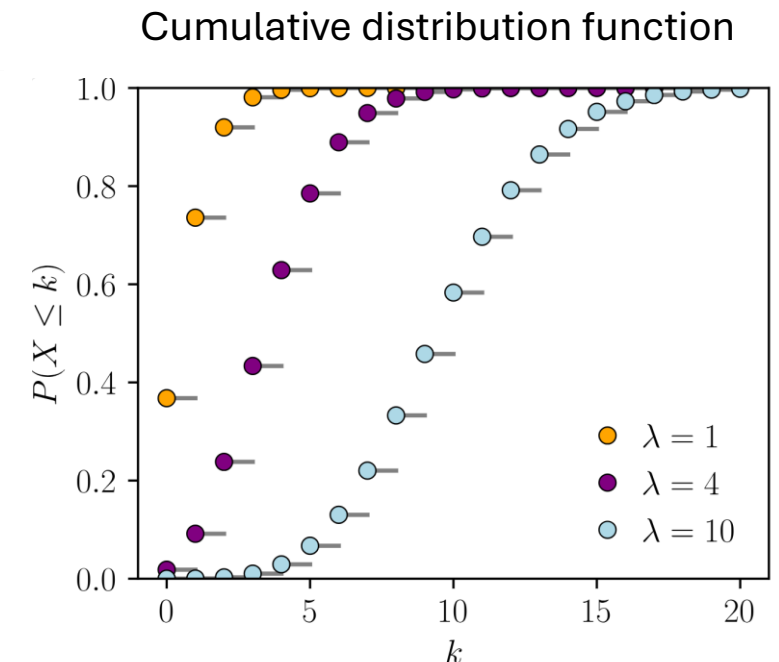
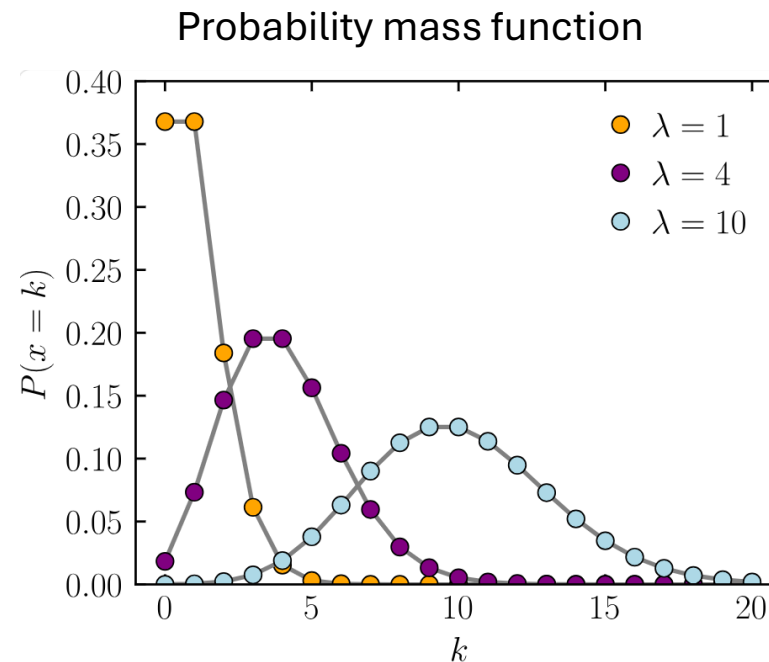
- Example for binomial distribution
- Action potential → calcium influx → vesicles with neurotransmitter fuse with membrane
- Probabilistic release of each quantum of transmitter
- Probability of x quanta being released (Katz, 1969)

$$P(x) = \binom{n}{x} p^x (1 - p)^{n-x}$$



Poisson distribution

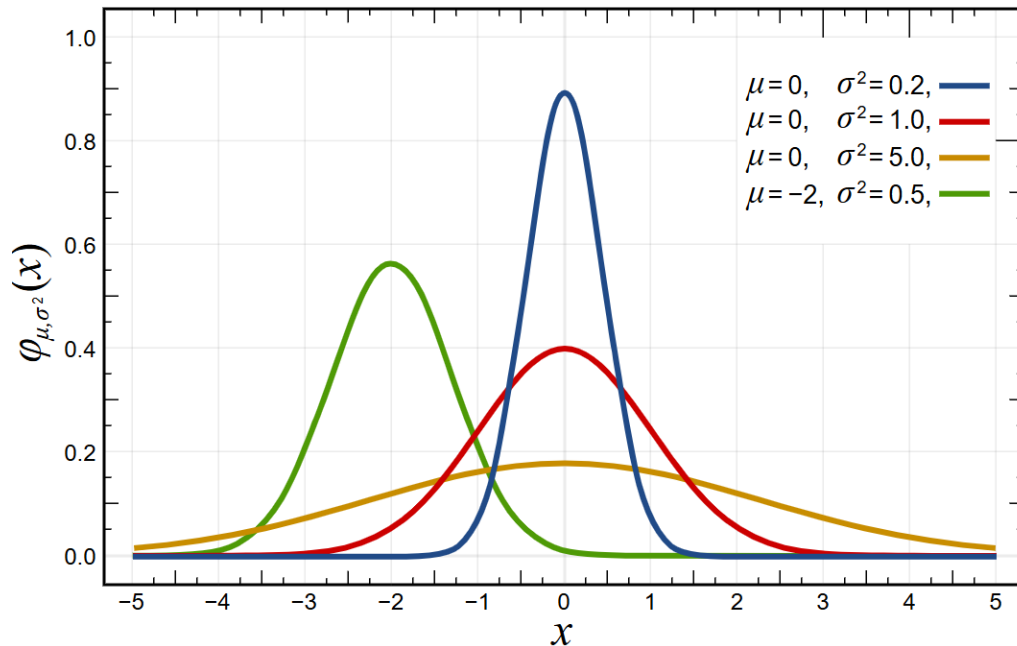
- Used to describe counts of binary events
- Assigns probability to every nonnegative integer
- $P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$
- $E(X) = \lambda$
- $\text{Var}(X) = \lambda$



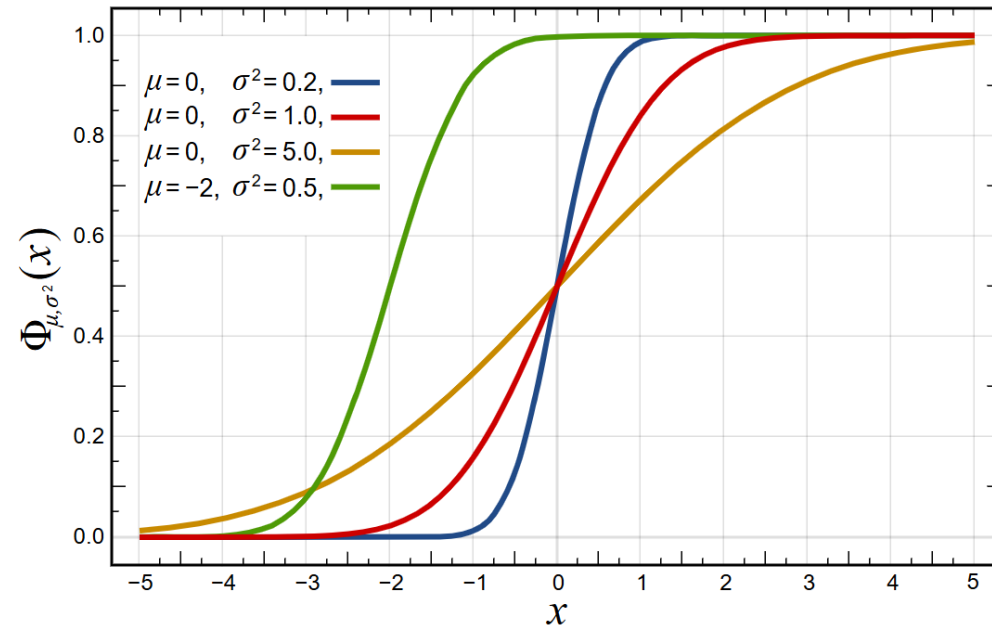
Normal distribution (Gaussian distribution)

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Probability mass function



Cumulative distribution function



Spike train

- Discrete series of time events at which action potential is generated
- Stochastic
- Irregular discharge pattern of neuron

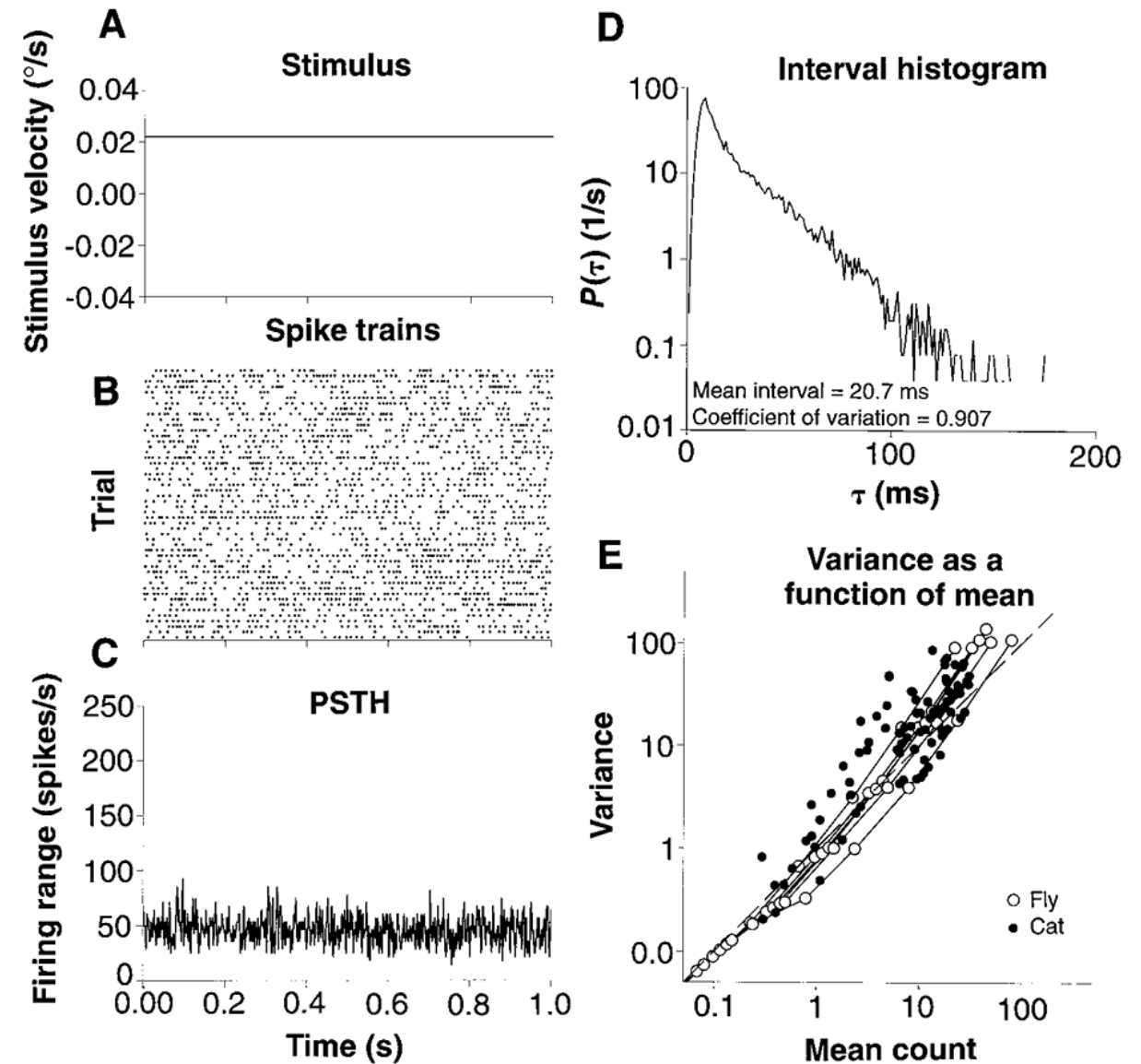
What property encodes information?

- Mean rate code (firing frequency averaged over long time)
- Intervals between spikes

Spike train variability

- Spike train of 50 trials
 - Spike rate (3 ms window)
 - Inter-spike interval (ISI)
 - Spike count
- High variability

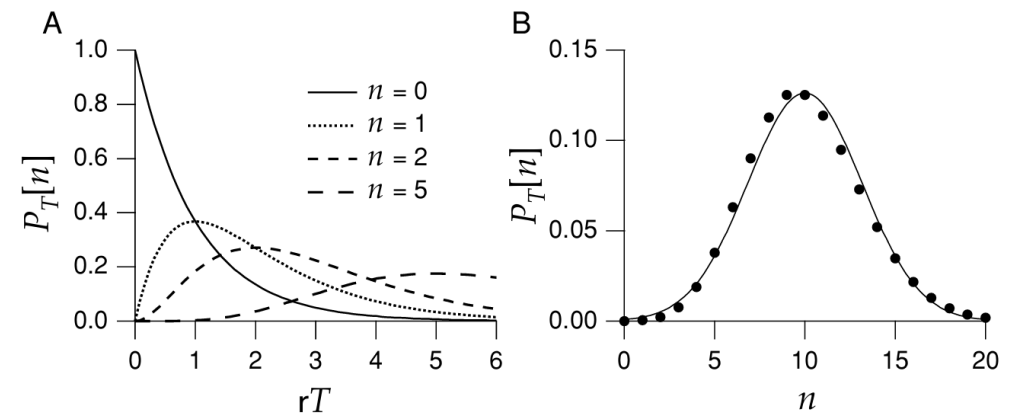
We need to work with probabilities



Poisson Process

- Extremely useful to approximate stochastic neuronal firing
- Describes a stochastic process that generates events
- In our case: Spikes
- Gives probability of a spike train of n spikes within a given time interval T and a time-dependent firing rate r

$$p[t_1, t_2, \dots, t_n] = \exp\left(-\int_0^T dt r(t)\right) \prod_{i=1}^n r(t_i)$$



Dayan and Abbott, Theoretical Neuroscience

Poisson Spike Generator

- Spike sequence is generated by estimate of firing rate $r_{est}(t)$
- Needed: probability of spike during short time interval
- Iterate over time steps Δt
- Generate random number x_{rand}
- Compare $r_{est}(t)\Delta t$ with x_{rand}
- Spike if $x_{rand} < r_{est}(t) \Delta t$

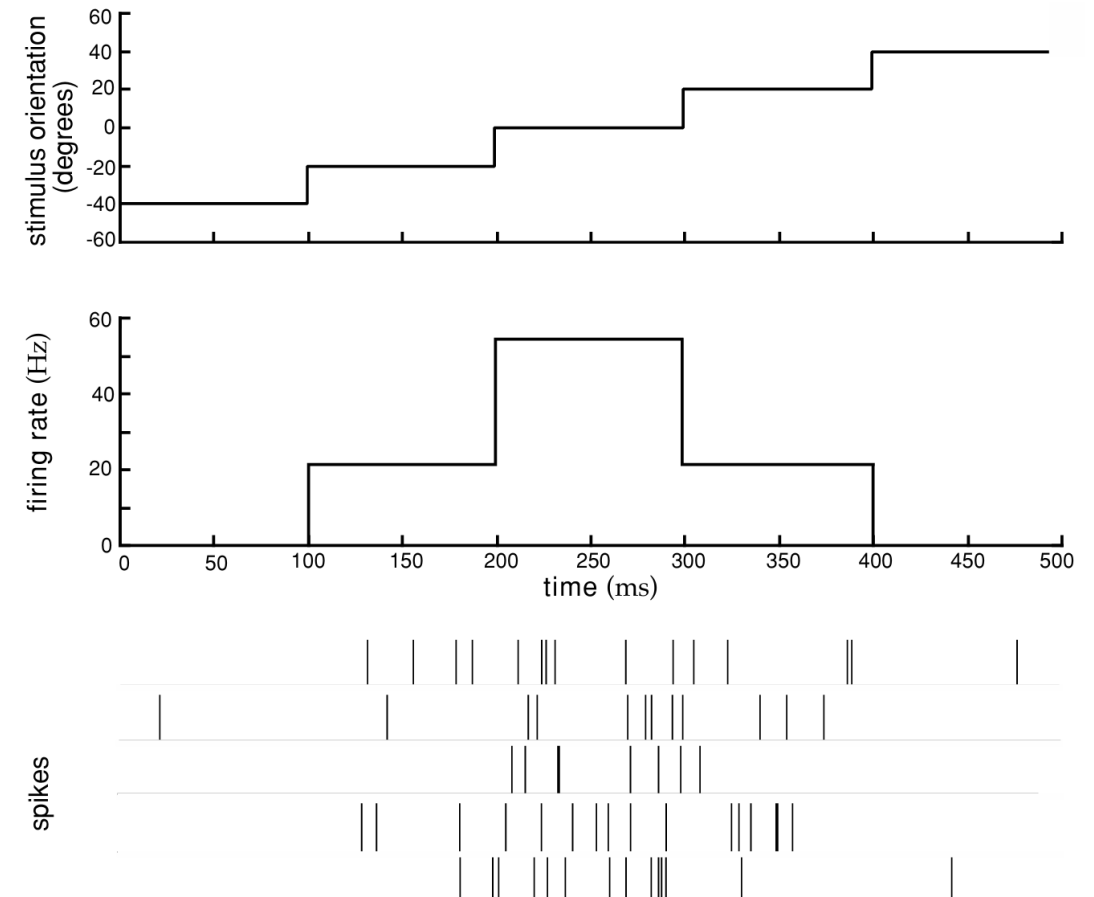


Figure 1.13 Model of an orientation-selective neuron. The orientation angle (top panel) was increased from an initial value of -40° by 20° every 100 ms. The firing rate (middle panel) was used to generate spikes (bottom panel) using a Poisson spike generator. The bottom panel shows spike sequences generated on five different trials.

Spike Count

- Fano factor describes relationship between mean spike count and variance
- $\sigma_n^2 = A \bar{n}^B$
- Experiments: A and B between [1,1.5]
- Poisson model: A=B=1
- Fano factors fitted with experimental data show a higher variability than Poisson models would predict

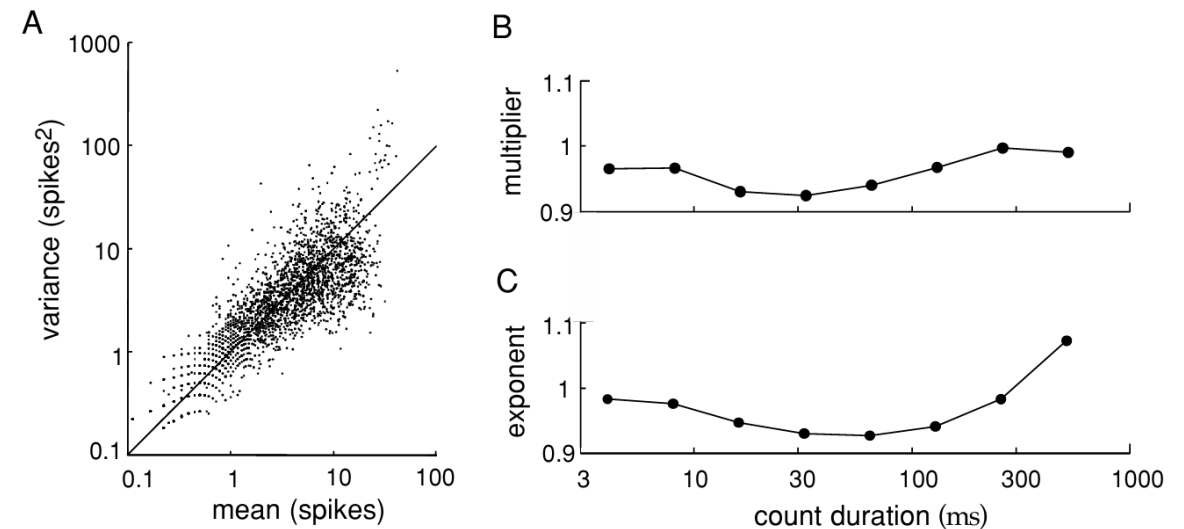
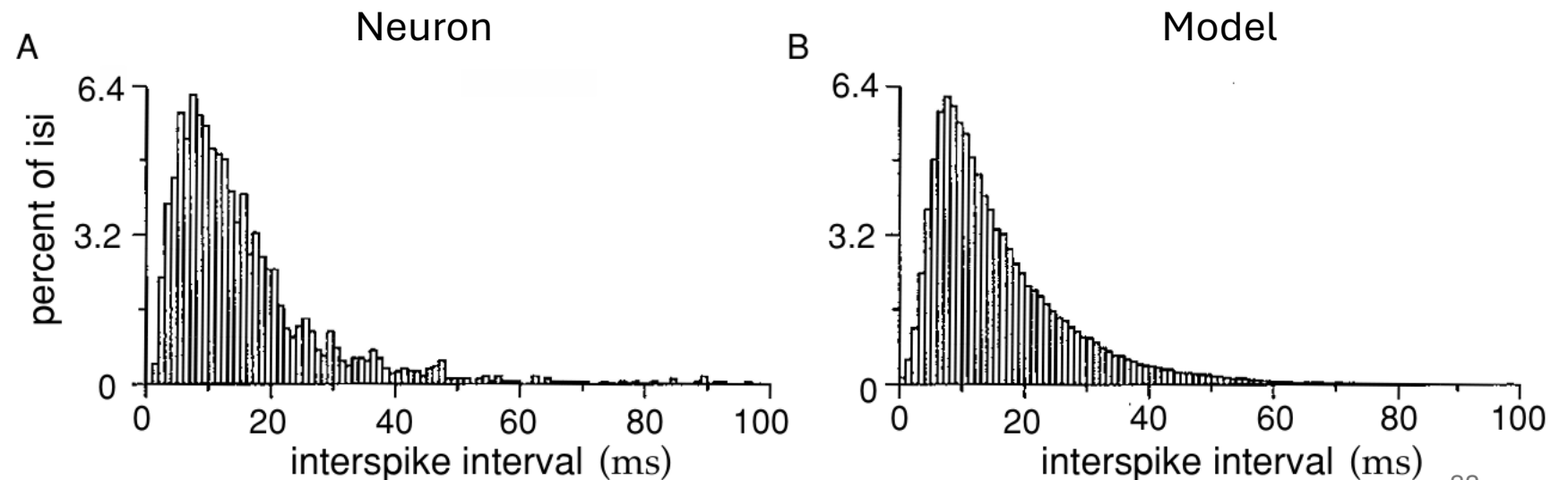


Figure 1.14 Variability of MT neurons in alert macaque monkeys responding to moving visual images. (A) Variance of the spike counts for a 256 ms counting period plotted against the mean spike count. The straight line is the prediction of the Poisson model. Data are from 94 cells recorded under a variety of stimulus conditions. (B) The multiplier A in the relationship between spike-count variance and mean as a function of the duration of the counting interval. (C) The exponent B in this relation as a function of the duration of the counting interval. (Adapted from O'Keefe et al., 1997.)

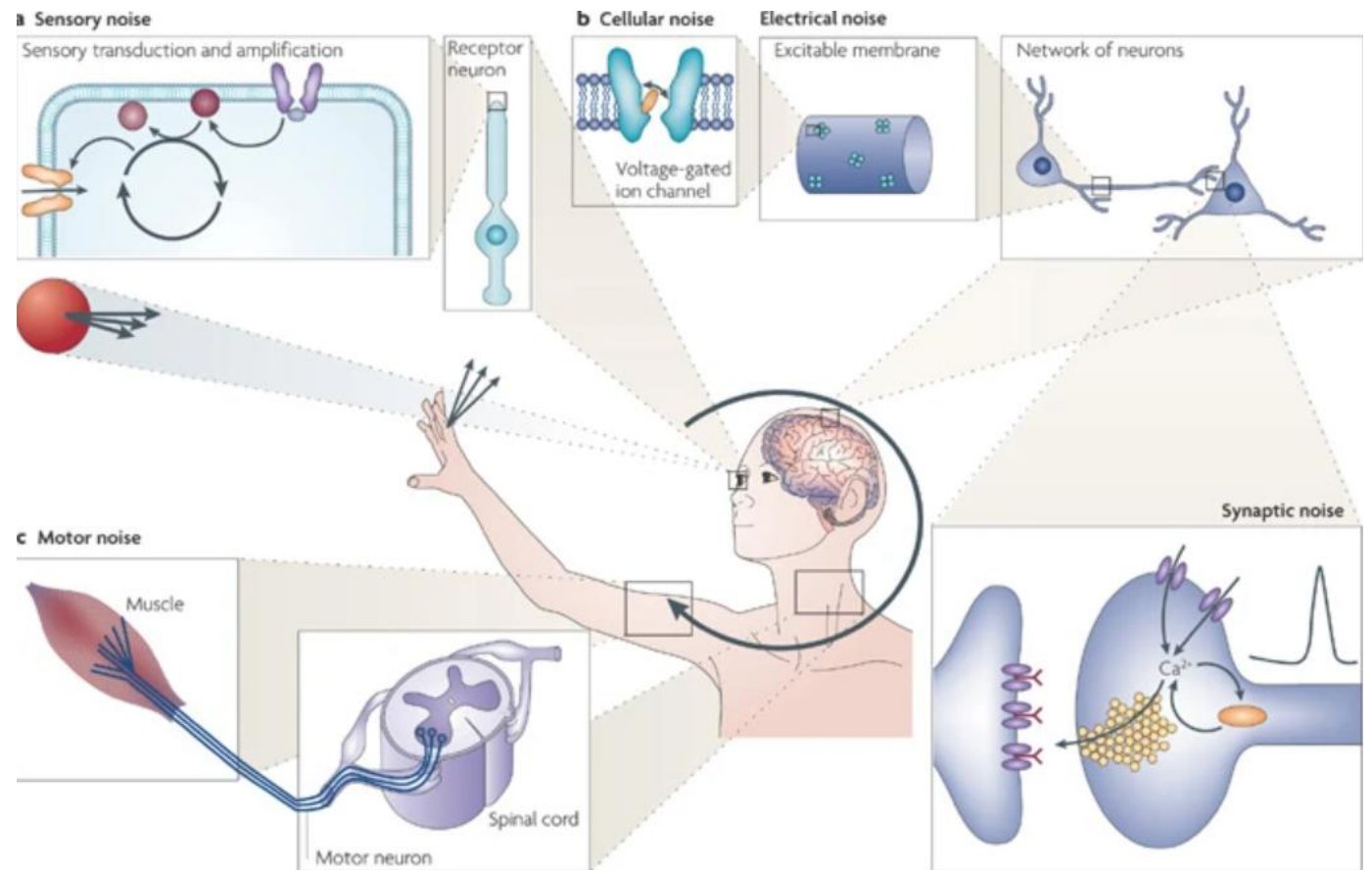
Interspike interval (ISI)

- Neuron recorded from cortical area responding to a moving image
- ISI calculated from spike train
- Counting number of intervals falling in discrete time bins
- Poisson model with stochastic refractory period



Noise

- Thermal motion of molecules (“Intrinsic noise”)
- Network noise
- Measurement noise



Network Noise

- “Extrinsic noise”
- Neurons receive input from thousands of other neurons
- A lot of presynaptic spikes do not generate a postsynaptic response
- High irregular spike trains
- Can be modeled by adding noise term to differential equation

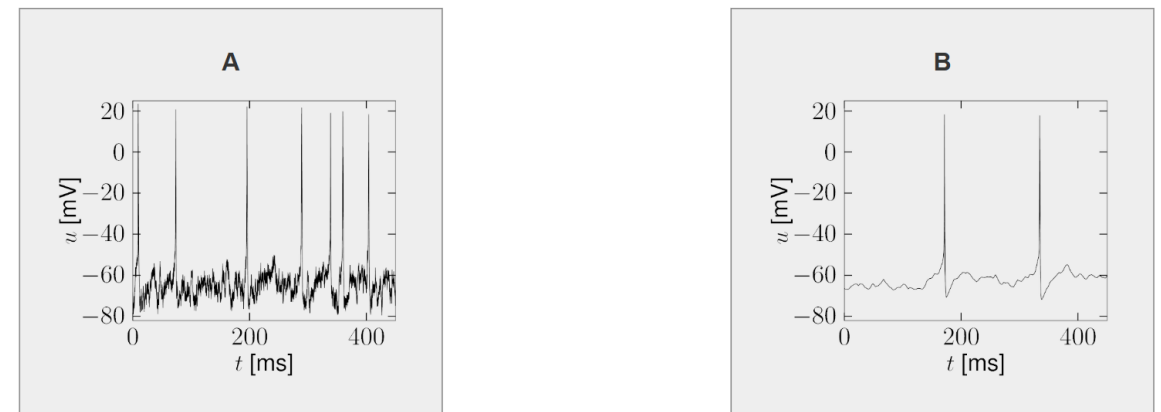


Fig. 8.1: Noisy input. **A.** A Hodgkin-Huxley model with parameters as in Chapter 2 driven with white-noise input. **B.** The same model driven with colored noise with time constant $\tau_s = 50\text{ms}$. Note that the fluctuations of the membrane potential are slower.

Network Noise

$$\langle \xi(t) \rangle = 0,$$

$$\langle \xi(t) \xi(t') \rangle = \sigma^2 \tau_m \delta(t - t')$$

White noise

- Stochastic process
- Expected value is 0
- Standard deviation is amplitude
- Added to differential equation

$$\tau_m \frac{d}{dt} u(t) = f(u(t)) + R I^{\text{det}}(t) + \xi(t)$$

Colored noise

- Filtered white noise

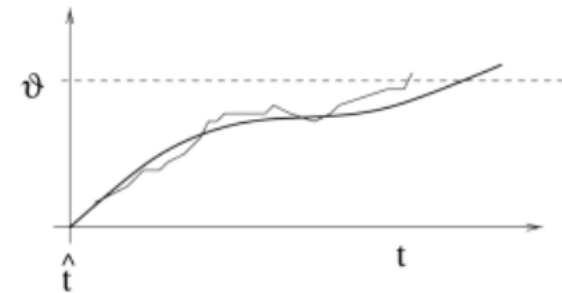


Fig. 8.2: Noisy integration. A stochastic contribution in the input current of an integrate-and-fire neuron causes the membrane potential to drift away from the reference trajectory (thick solid line). The neuron fires if the noisy trajectory (thin line) hits the threshold ϑ .

Measurement Noise

- Measurement artifacts also lead to trial to trial variability
- Background noise
- Biological Noise (EMG, EOG, EEG, ...)
- Powerline noise
- Noise from technical equipment (amplifier, electrodes)

→ Filter

→ Repeat measurement and average over trials

→ Probabilistic modeling

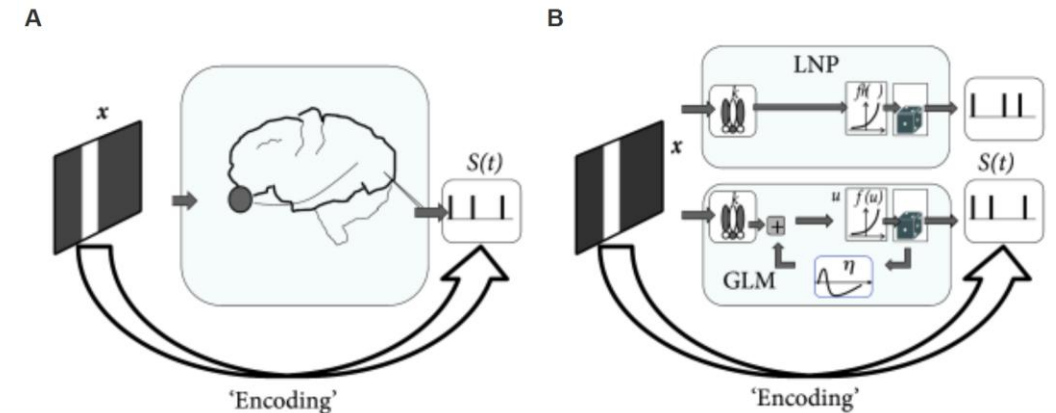
Neural data analysis

Encoding

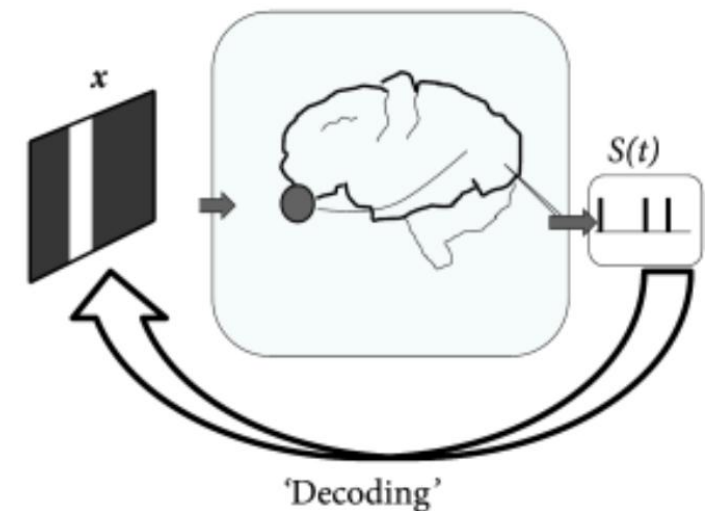
- How information is encoded in neural spike trains
- Predict spike trains of a neuron
- Given synaptic input, current injection, or sensory stimulus

Decoding

- What can we learn from observation of spike sequence
- Estimate stimulus that gave rise to spike train



Gerstner et al., Neural Dynamics



Gerstner et al., Neural Dynamics

Model selection

- Flexible and powerful enough to fit observed data
- Tractable (possible to fit model with available data)
- Respect underlying physiology and anatomy of system
- Biophysical interpretability

Statistical Formulation

- D ... spike train data
- n_t ... variables of discretized spike train
- T ... time of spike train
- θ ... set of all model parameters

Encoding Model

- Assigns conditional probability to any possible neural response D given stimulus $x \rightarrow P(D|x)$
- Not possible to directly measure p for all pairs x, D because there are infinitely many potential stimuli $\rightarrow P(D|x, \theta)$
- Goal: Estimate model parameters θ so that model fits the observed data D
- With estimated θ we can then compute response probability for new stimuli x

Parameter Estimation

- Compute posterior distribution of model parameters given observed data using Bayes` rule
- Posterior: $P(\theta|D)$
- Likelihood: $P(D|X, \theta)$
- θ ... model parameter
- X ... set of stimuli
- Find θ that maximizes likelihood or posterior

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(\theta|D) \approx P(D|\theta) * P(\theta)$$

$$\hat{\theta}_{ML} = \arg \max_{\theta} P(D|X, \theta)$$

$$\hat{\theta}_{MAP} = \arg \max_{\theta} P(D|X, \theta) * P(\theta)$$

Likelihood $P(D|X, \theta)$

- Gives us probability of observing the data D given stimulus X and parameter θ
- We want to find the parameter θ , which gives us the highest probability of observing our data D
- Maximize Likelihood function

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(\theta|D) \approx P(D|\theta) * P(\theta)$$

$$\hat{\theta}_{ML} = \arg \max_{\theta} P(D|\theta)$$

$$\hat{\theta}_{MAP} = \arg \max_{\theta} P(D|\theta) * P(\theta)$$

Uncertainty

- A probabilistic model predicts a **distribution of possible responses**, not necessarily one response
- Use confidence intervals to give a range of plausible parameter values
- Use mean squared error to estimate accuracy
- Bias: Systematic error
- Variance: High variability of estimate

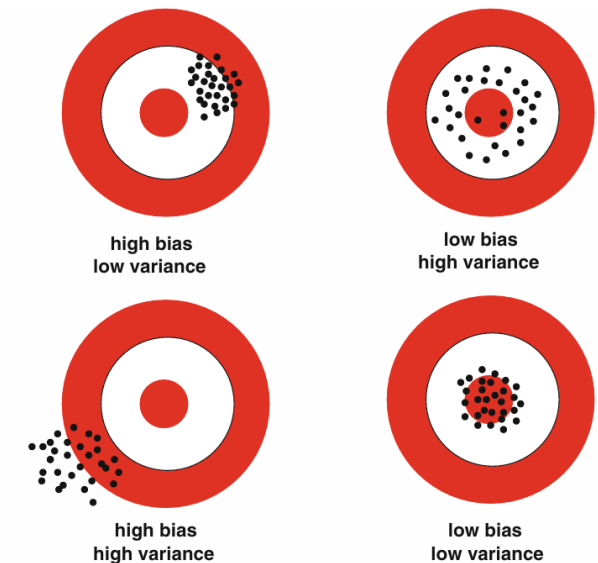


Fig. 8.1 Drawing of *shots* aimed at a target to illustrate the way estimates can miss their “target.” They may be systematically biased, or they may have high variability, or both. The best situation, of course, is when there is little systematic bias and little variability.

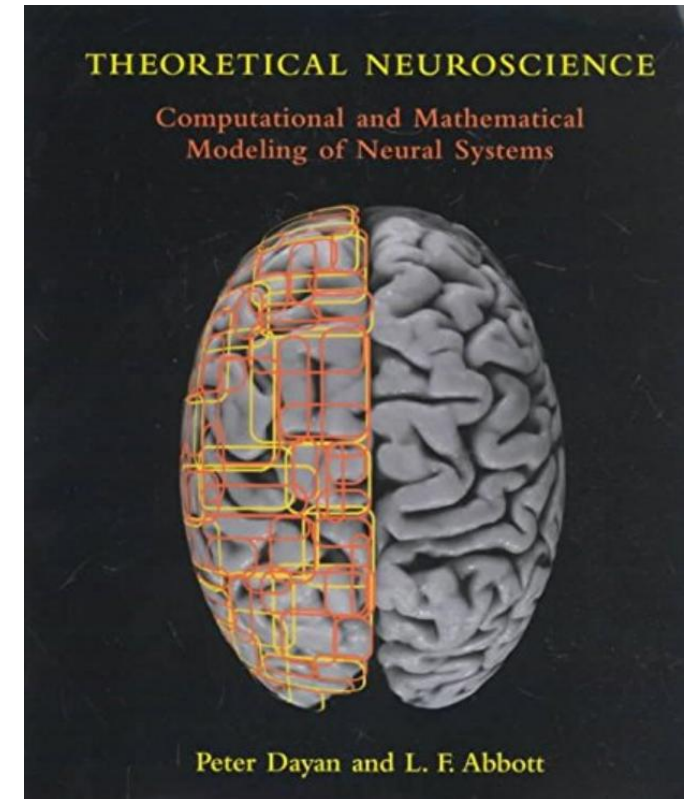
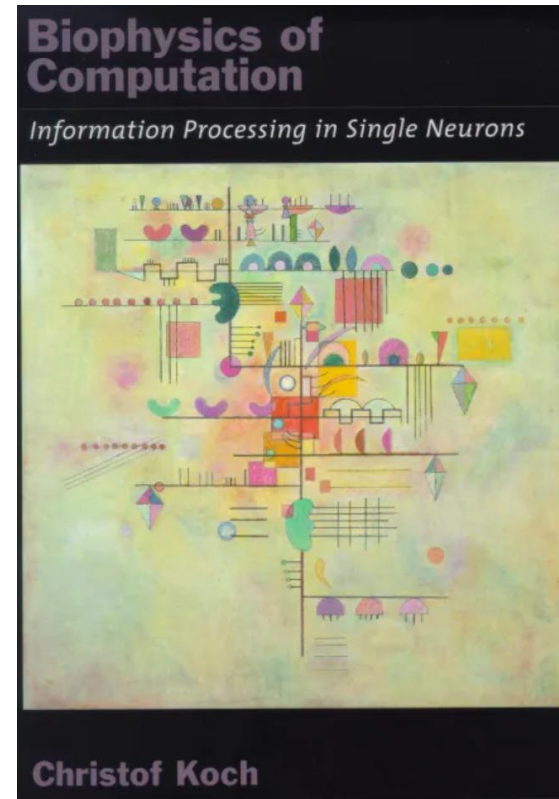
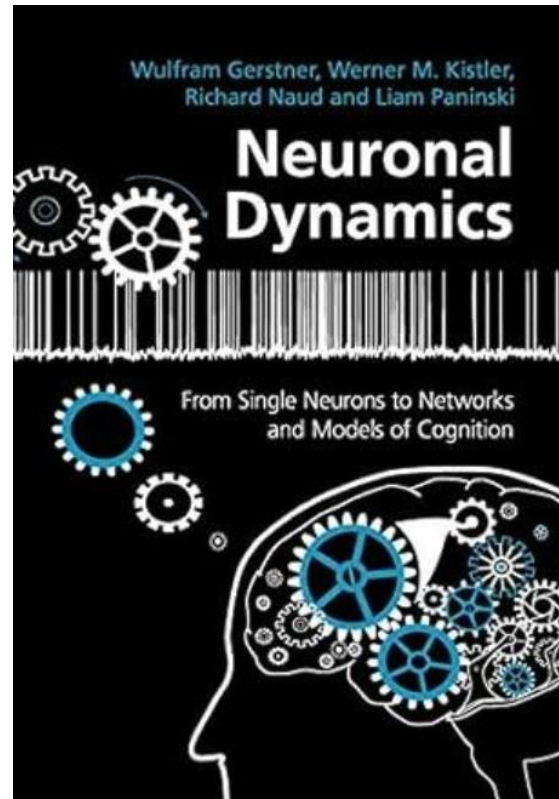
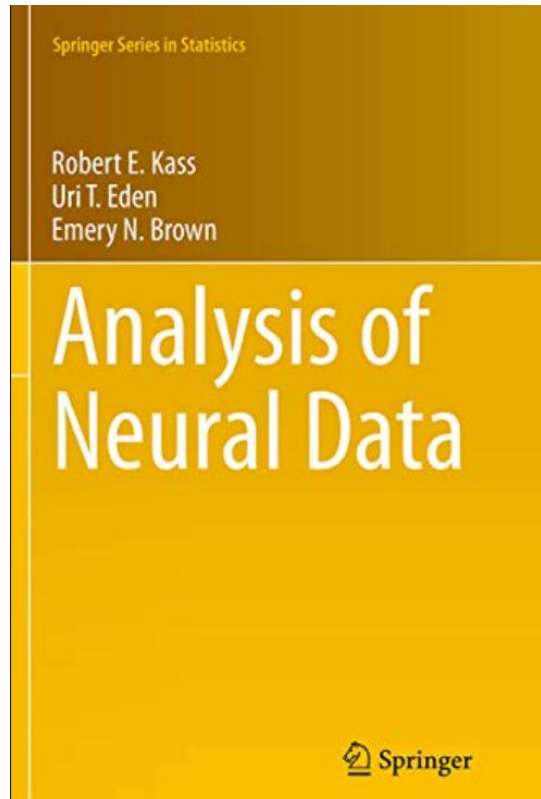
To sum up

Probability gives us a tool to

- Understand variability in neural systems
- Simulate stochastic spike trains
- Fit stochastic models
- Evaluate estimated parameters and models

If you want to know more about ...

... Computational Neuroscience



... single particles and thermal motion

Leonard Susskind: Lecture on “Statistical Mechanics” on YouTube

(not neuroscience related but very interesting to understand concept of thermal motion and where the “noise” comes from)